Aligned but Inert: Concept Bottlenecks in Dense Prediction, and How to Make Them Causal

Anonymous WACV Applications Track submission

Paper ID 2592

Abstract

Concept bottleneck models promise intrinsic interpretability: route the prediction through a human-readable concept layer, and the concept activation is *the explanation. We show that transplanting this idea into dense prediction fails silently, diagnose why, and repair it. We build a concept-routing bottleneck that assigns every encoder token to one of K clinically-labeled experts, and train it for 3D brain-tumor segmentation. By every measure this literature reports, it succeeds: routing tracks the clinical concepts at* $\mathrm{CAS}_{\mathrm{fg}}$ *up to* 0.96 *and separates necrosis, edema and enhancing tumor at AUROC* 0.88*, against* 0.74 *for the strongest post-hoc attribution. Yet the routing is* causally inert*. Forcing tokens through a different expert changes* 0.2% *of voxel labels, and zeroing the entire bottleneck costs* 1.5 *Dice. Alignment metrics cannot detect this, because alignment is compatible with a bottleneck the model never consults—and the alignment loss trains precisely what those metrics score. We trace the failure to two leaks that encoder–decoder architectures invite, a residual path* through *the module and skip connections* around *it, and rule out the obvious explanation: the experts are not degenerate (*0.9 *pairwise divergence). A linear probe on an encoder trained with no concept supervision already reaches* 0.81*, so concept-discriminability gains over saliency baselines are weaker evidence than they appear. Attempts to force dependence by* removing *the decoder's alternatives fail—the optimizer compensates. Making the routing an* additive term of the output *instead succeeds: causal control rises from* 0.011 *to* $\mathbf{0.636}$ $[0.621, 0.651]$ *label-flip under intervention (*$p{<}10^{-28}$*), removing the routing term costs up to* 99% *Dice, and the price is* ≈ 1 *Dice point.* ***A concept layer is an explanation only if the model consults it, and showing that requires intervening, not correlating.***

1. Introduction

Deep segmentation models match expert radiologists on brain-tumor benchmarks, yet clinical deployment lags because their decisions cannot be verifiably explained. The dominant remedy is post-hoc attribution [14, 19, 21], which produces a saliency map *after* the prediction and therefore approximates the mechanism rather than reporting it. Rudin [18] argues for the alternative: build models whose decisions are interpretable by construction. Concept Bottleneck Models (CBMs) [12] realize this for classification by forcing the prediction through a human-readable concept layer, so that the concept activation *is* the explanation.

Extending that idea to dense prediction is attractive, and—as we show—treacherous. We build the natural extension: replace the encoder bottleneck of a segmentation network with a router that assigns every token to one of K clinically labeled experts {Background, Necrotic core, Edema, Enhancing tumor}. The routing distribution is a per-region, concept-labeled map computed in a single forward pass with no post-hoc step. It looks exactly like an intrinsic explanation should.

It aligns beautifully, and it does nothing. Trained end-to-end, the routing tracks the clinical concepts closely ($\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96) and discriminates them far better than any post-hoc method (0.88 vs. 0.74 for the strongest baseline). These are the metrics this literature reports, and by both the module is a success. But when we *intervene*—force the tokens the router assigned to one concept through a different concept's expert, changing nothing else—the prediction does not move. Only 1.1% of affected voxels change label, and zeroing the entire bottleneck costs 1.5 Dice on edema.

The failure is invisible to alignment metrics *by construction*. $\mathrm{CAS}_{\mathrm{fg}}$ measures correlation between the routing and the ground truth, and correlation is perfectly compatible with a bottleneck the model never consults. Worse, the alignment loss trains precisely what $\mathrm{CAS}_{\mathrm{fg}}$ scores, so a high value confirms that the router learned its supervision—not that the network uses it.

Why it happens. We rule out the obvious explanation: the experts have *not* collapsed to a shared function. They displace tokens by 59–74% of their norm and differ pairwise by 0.93. The cause is architectural, and it is two independent leaks that encoder–decoder designs invite. The expert computes $h + c_k(h)$, so the residual carries the full feature vector to the decoder whichever expert fires—routing modulates the representation but never gates it. And the decoder's skip connections carry encoder features directly to the output, so it can decide what tissue it is looking at without consulting the routing. Either leak alone suffices, and neither is visible in $\mathrm{CAS}_{\mathrm{fg}}$.

A control that reframes the baseline. We also find that a linear probe on an encoder trained with *no* concept supervision already separates the concepts at 0.81—above every post-hoc method we test. Concept-discriminability gains over saliency baselines are therefore much weaker evidence for a concept module than the literature treats them as.

Closing the leaks. Attempts to force dependence by *removing* the decoder's alternatives fail: a scalar gate on the skip connections is compensated away during training, and removing the expert residual backfires, collapsing pairwise expert divergence from 0.93 to 0.16 because every expert must then independently reconstruct the full signal. Making the routing an *additive term of the output* succeeds, because causal influence becomes arithmetic rather than something the optimizer can route around. Causal control rises from 0.011 to 0.636, removing the routing term costs up to 99% Dice, and the price is ≈1 Dice point.

Contributions.

1. **A diagnostic protocol for concept bottlenecks** (Sec. 4): causal intervention, bottleneck ablation, and a linear-probe control. Three cheap tests that separate a bottleneck the model uses from one it ignores. Alignment metrics cannot.
2. **Evidence that the natural design fails it** (Sec. 5.3). A bottleneck with $\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96 and best-in-class concept discriminability is causally inert, and we localize the cause to two specific architectural leaks.
3. **A repair and its price** (Sec. 5.6). Additive routing makes the bottleneck causally decisive—0.636 $[0.621, 0.651]$ label-flip under intervention against 0.011 $[0.009, 0.014]$, $p < 10^{-28}$—for ≈1 Dice point, with alignment and calibration slightly improved.

2. Related Work

Post-hoc attribution. Grad-CAM [19], integrated gradients [21], SHAP [14] and attention rollout [1] reconstruct a saliency map after the prediction. Their reliability has itself been questioned: popular methods can fail basic sanity checks [2], and the analogous audit found attention weights a poor guide to what a model used [10]. We apply that tradition to intrinsic rather than post-hoc explanation. On dense prediction the extension proceeds through heuristics, and the output is an input-*sensitivity* map rather than a concept assignment. We use four such methods as baselines under an identical protocol. Their central limitation for our purposes is structural rather than empirical: a saliency map is a readout, so there is nothing in it to intervene on. The causal test in Sec. 4 simply cannot be run on them.

Intrinsic interpretability and concept bottlenecks. CBMs [12] and ProtoPNet [5] make the explanation part of the computation, for scalar-output classification. Concepts have also been probed in networks not built around them [4, 11]; those methods measure alignment between an internal signal and a concept, precisely the quantity we show is insufficient. Neither addresses dense prediction, where the explanation must be per-voxel. Critically for this paper, the standard validation in this line is *concept accuracy or alignment*—does the concept layer predict the annotated Reported failure modes for CBMs concern what the concept vector *encodes*—whether it carries task information beyond the named concepts. Ours is complementary and more basic: the concept vector may encode the concepts perfectly and still not be read, because the information bypasses the bottleneck through the *architecture* rather than through the concept encoding.

Mixture-of-experts routing. Sparse MoE [7, 16, 20] introduced routing for conditional computation. There, routing is causally decisive because the selected expert's output *is* the layer output. Our results show that transplanting the mechanism into a skip-connected encoder–decoder silently voids that property—which is exactly the assumption an interpretability claim would rest on. We inherited the intuition without checking that the conditions supporting it still held.

3. Concept Routing at the Bottleneck

3.1. Router, experts, dispatch

Fig. 1 gives the module and the two leaks at a glance. Let $h_t \in \mathbb{R}^D$ be the bottleneck token at spatial position t. A router produces

$$p(t) = \mathrm{softmax}\big(W h_t/\tau + \beta\,\mathrm{sim}(h_t, \Pi)\big) \in \Delta^{K-1}, \quad (1)$$

where Π are K learnable concept prototypes and τ an annealed temperature. Each concept owns an expert E_k.

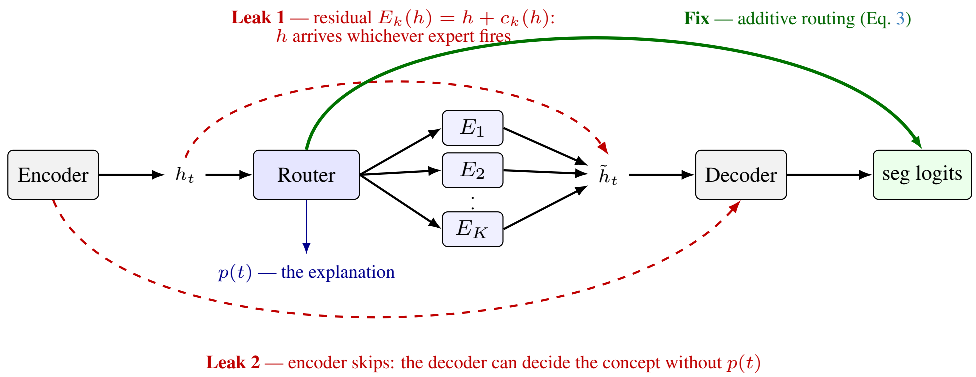

Figure 1. **Concept routing, and the two paths that make it inert.** The router assigns each bottleneck token h_t to one of K clinically labeled experts; the routing distribution $p(t)$ is read from the forward pass and is the explanation. The design is intrinsic only if $\tilde{h}_t$ is the decoder's *sole* source of concept information, and two standard choices break that: the expert residual delivers h intact whichever expert fires, and encoder skips let the decoder decide the concept without consulting $p(t)$. Either alone makes the routing inert, and neither is visible in any alignment metric. The fix adds the routing logits to the output rather than removing the decoder's alternatives, which the optimizer absorbs (Sec. 3.3).

Training uses soft dispatch, inference hard dispatch:

$$\tilde{h}_t = \sum_k p_k(t) E_k(h_t) \quad \text{(soft)}, \qquad \tilde{h}_t = E_{k^*(t)}(h_t) \quad \text{(hard)}, \tag{2}$$

where $k^*(t) = \arg\max_k p_k(t)$ selects the highest-probability expert. $p(t)$ is the explanation: a per-region distribution over clinical concepts, read directly from the forward pass. We train with $\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{div}} + \lambda_3 \mathcal{L}_{\text{ent}} + \lambda_4 \mathcal{L}_{\text{bnd}}$, where $\mathcal{L}_{\text{align}}$ is a foreground-only focal BCE pushing $p_k(t)$ toward the ground-truth concept indicator, under a three-phase curriculum (warm-up / alignment / refinement) that anneals τ from 2.0 to 0.5.

Note already that $\mathcal{L}_{\text{div}}$ acts on routing *vectors*, not on expert *functions*: nothing in the objective forces $E_a \neq E_b$. We therefore verify functional divergence empirically rather than assume it (Sec. 5.3).

3.2. Two leaks

Eq. 2 yields an intrinsic explanation *only if* $\tilde{h}_t$ is the decoder's sole source of concept information. Two standard design choices break that.

Residual (through the module). The natural expert is a pre-norm residual MLP, $E_k(h) = h + c_k(h)$, whose residual stabilizes early training. It also means the decoder receives h intact regardless of k^*: routing reweights a signal that arrives anyway.

Skips (around the module). Encoder–decoder segmentation networks connect encoder stages directly to matching decoder stages [17], a design so standard that it is the default in the strongest current medical segmentation systems [9]. Every such connection is a path from input to output that does not pass the bottleneck. A related placement issue is easy to miss: in a hierarchical transformer encoder the "bottleneck" is ambiguous, and inserting the module at an intermediate stage leaves *deeper*, more semantic stages flowing around it. We report this because our own first design made exactly that error.

3.3. Three ways to close them

We expose all three as switches, and evaluate each.

(i) Gated expert. $E_k(h) = c_k(h)$, so the expert transform is the only path. The standard near-zero initialization of c_k's output layer must be relaxed here, since c_k is now the signal rather than a correction.

(ii) Skip attenuation. A scalar α scales the bypassing skips, sweeping from α=1 (standard) to α=0 (bottleneck is the only route). Skips carry spatial detail as well as semantics, so α trades boundary quality against causal control.

(iii) Additive routing. Rather than removing the decoder's alternatives, make the routing a term of the output:

$$\text{seg\_logits} = \text{up}\big(\ell(t)\big) + \lambda \cdot \text{decoder\_out}, \tag{3}$$

where $\ell(t)$ are the router logits upsampled to voxel resolution and λ scales a bounded refinement. The routing supplies the coarse per-concept prediction; the decoder keeps every skip and contributes detail—which is what skips are good at.

Options (i) and (ii) are *subtractive*: they fight an architecture that segments perfectly well without its bottleneck. Option (iii) is *additive*, and the distinction turns out to be decisive. Under Eq. 3 causal influence is arithmetic: at inference the refinement is a function of decoder features, so intervening on the routing shifts the output by exactly

$\mathrm{up}(\Delta\ell)$. No weight configuration undoes an additive term after the fact.

The corresponding failure mode is magnitude rather than mechanism: the network can shrink the routing term *relative* to the refinement, leaving the effect arithmetically real and numerically negligible, since $\mathcal{L}_{\text{align}}$ pins the routing's direction but not its scale. We therefore monitor $\rho = \|\lambda \cdot \text{refine}\|/\|\mathrm{up}(\ell)\|$ throughout training rather than assuming it stays small.

3.4. Intervening on a two-path module

One implementation detail is essential and easy to get wrong. Under Eq. 3 the routing reaches the output through *two* paths: the expert features the decoder consumes, and the additive term. An intervention that overrides only the dispatch leaves the additive path reading the router's original logits, so it silently does nothing through the path that matters. We therefore expose an *effective* logit tensor that reflects the forced decision, and the model consumes that; the reported explanation remains the router's actual belief, so an intervention can never contaminate the map we publish.

The hardening must also be applied uniformly. If only the overridden tokens are converted to a one-hot, an identity intervention $a \to a$ swaps soft logits for a hard assignment and moves the prediction even though nothing was re-routed—in our case by 14.3 logits—which destroys the diagonal control that licenses reading the off-diagonal cells. Applying it to every token makes the baseline and every intervention commensurable, and restores $a \to a$ to a bit-exact no-op. We verified both properties before trusting any number in Sec. 5.6.

The same care applies to ablation. Patching only the decoder's bottleneck input leaves the additive term intact and measures a 0.09 logit change where the true effect is 7.0; a study using that measurement would conclude the bottleneck is unnecessary and reject a working design.

4. A Diagnostic Protocol

Alignment is necessary but not sufficient. We propose three tests, all cheap, that together establish whether a concept bottleneck is load-bearing.

Definitions. $\mathrm{CAS}_{\mathrm{fg}}(k) = \mathrm{Pearson}\big(p_k(t), M_k(t)\big)$ over foreground tokens measures routing/ground-truth *alignment*, where M_k is the ground-truth concept indicator. Concept-discriminability is the AUROC of p_k as a classifier for concept k *among the other tumor voxels*—a harder question than localization, which only asks tumor vs. brain.

Test 1: causal intervention. Replace $k^*(t)$ with a chosen b for the tokens assigned to a, changing nothing else, and re-run the decoder. Report the change in the posterior of class b over exactly those voxels, the fraction of their labels that flip to b, the change in *off-target* classes (specificity), and the change *outside* the intervened region (locality).

Three details make this a control rather than a demonstration. The baseline must run through the same hard-dispatch path, so the identity intervention $a \to a$ is a bit-exact no-op and any off-diagonal effect is attributable to re-routing alone. When a model consumes routing through more than one path—as Eq. 3 does—the intervention must move *all* of them, or it silently does nothing through the one it missed. And the intervention should be restricted to the tumor neighborhood: a router typically sends most of the volume to background, so re-routing every token of a concept repaints the whole brain, which demonstrates that the output follows the routing everywhere rather than concept control where it is clinically meaningful. We report both scopes and quote the restricted one.

This test is impossible for post-hoc attribution. There is nothing in a saliency map to intervene on, because the map is a readout: editing it changes nothing about the model.

Test 2: bottleneck ablation. Replace the module output with the raw encoder features (*bypass*) or with zeros (*zero*), and measure the Dice change. If destroying the bottleneck barely hurts, the decision does not live there, and no amount of concept alignment will make it interpretable. When the module reaches the output through two paths, each must be ablated separately: patching only the decoder's bottleneck input leaves an additive routing term intact and wrongly reports the bottleneck as unnecessary.

Test 3: linear-probe control. Fit multinomial logistic regression on the features *entering* the module and score it exactly as the router is scored. This asks whether the concept structure is produced by the concept machinery or merely read off features that already contained it. Run it also on a model trained without concept supervision, for the strong version of the control.

5. Experiments

5.1. Setup

BraTS 2024 glioma [3, 15]: four co-registered MRI modalities, labels remapped to {0:Background, 1:NCR, 2:Edema, 3:ET}, patient-wise 75/12.5/12.5 split (n_{test}=168), z-scored per modality, 128^3 inputs, 80 epochs of AdamW (10^{-4}, cosine), mixed precision. The bottleneck is 192×16^3. We report a SegResNet backbone throughout for comparability across configurations; a Swin transformer with a UNETR decoder reproduces every qualitative finding. Three configurations recur:

V0 — the natural design (residual expert, skips intact).

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Table 1. The natural design by standard metrics (BraTS test, n=168). Concept-discrim. is concept k vs. the *other* tumor voxels; 0.5 is chance.

	Dice WT/TC/ET	$\mathrm{CAS_{fg}}$ N/E/T	Concept-discrim.
V0	0.93/0.88/0.87	0.80/0.96/0.95	**0.883**
Occlusion	—	—	0.741
Gradient	—	—	0.601
Int. grad.	—	—	0.583
Grad-CAM	—	—	0.521

V2 — subtractive repair: gated expert, α=0.5.
DIRECT — additive routing, Eq. 3, λ=1.

5.2. The natural design aligns

Table 1 reports the standard picture. Segmentation is competitive, routing tracks the concepts closely, and on concept-discriminability V0 reaches 0.88 against 0.74 for the strongest post-hoc method, with integrated gradients and Grad-CAM near chance. By every measure this literature reports, the module works.

Removing $\mathcal{L}_{\text{align}}$ collapses $\mathrm{CAS_{fg}}$ to $-0.01/0.32/0.09$ while Dice barely moves (0.905 vs. 0.927 WT). We read this at the time as evidence that alignment is *learned* rather than incidental. It is—but the same number says something we missed: the segmentation does not need the alignment.

5.3. . . . and is inert

Intervention. Re-routing moves almost nothing. Averaged over concept pairs, 1.1% of re-routed voxels change label (Table 2, V2 row; V0 gives 0.1% on a smaller pilot). The identity diagonal is exactly 0, so this is a property of the model rather than of the instrumentation.

The experts are not degenerate. The obvious explanation is false. Experts displace tokens by 0.59–0.74 of their norm and differ pairwise by 0.93 (relative L_2). The routing selects genuinely different transformations; they simply do not reach the output.

Bottleneck ablation. They do not reach the output because the output does not need them. Replacing the module with raw features or with *zeros* costs almost nothing: edema Dice $0.879 \rightarrow 0.865$, ET $0.862 \rightarrow 0.806$. A decoder that can segment without its bottleneck will ignore any concept structure placed there.

5.4. The probe control

A linear probe on the features entering the module scores 0.839 against the router's 0.872. Run on a model trained with *no* concept supervision, the probe still reaches **0.811** —above every post-hoc method in Table 1. The concept structure is largely a property of a competent segmentation encoder, not of the concept machinery. This has a consequence beyond our system: reporting that a concept module separates concepts better than saliency baselines is weak evidence, because logistic regression on an unsupervised bottleneck clears that bar.

5.5. The output is a stronger concept classifier than the routing

A second control tightens the same point. The network's own segmentation posterior—free from the same forward pass—separates the concepts at 0.979, well above the routing's 0.883. It reaches 0.982 on the transformer backbone despite different pretraining and a different Dice, so the metric saturates for any competent segmenter: under this window it is close to a restatement of segmentation accuracy, because ranking voxels is far easier than thresholding them. Matched to the router's 16^3 decision budget by average-pooling and re-upsampling through the identical operator, the posterior falls to 0.935, so roughly half the residual gap is spatial resolution rather than concept information.

We report this because omitting it would overstate our case, and because it changes what a discriminability number means. The posterior is not an explanation: it *is* the prediction, and restating the output explains nothing about how the output arose. But together with the probe control (Sec. 5.4) it establishes that concept-discriminability cannot carry an interpretability claim on its own—an unsupervised linear probe clears the post-hoc baselines, and the model's own output beats everything.

5.6. Repair

Subtractive repair fails, instructively. Attenuating the skips (α=0.5) leaves every metric within noise of V0: Dice -0.002 WT, $\mathrm{CAS_{fg}}$ -0.001 on edema, and causal control still at 0.011. A scalar gate is compensable during training—normalization affine parameters can simply rescale to undo it—so at convergence it constrains nothing.

Removing the expert residual is worse than ineffective: pairwise expert divergence *collapses* from 0.93 to 0.16. With $E_k(h) = h + c_k(h)$ each expert supplies only a concept-specific correction and is free to differ; with $E_k(h) = c_k(h)$ every expert must independently reconstruct whatever the decoder needs, so they converge on approximately the same map. Making the expert the sole path made the experts interchangeable.

Additive routing works. Table 2 gives the result. Causal control rises from 0.011 $[0.009, 0.014]$ to **0.636** $[0.621, 0.651]$, a paired difference of $+0.625$

Table 2. Causal control and its price (BraTS test, n=168). Label-flip is the mean off-diagonal fraction of re-routed voxels whose predicted label became the target concept, tumor-restricted and scope-matched; the identity diagonal is exactly 0 in every configuration. Subtractive repair (V2) does not work; additive routing (DIRECT) does, for ≈1 Dice point.

	Repair	Label-flip ↑	95% CI	Dice WT/TC/ET	CAS_{fg} N/E/T	routing ECE ↓
V0	none (natural design)	0.001	(pilot, n=5)	0.927/0.883/0.869	0.80/0.96/0.95	0.841
V2	gated expert, α=0.5	0.011	[0.009, 0.014]	0.925/0.872/0.854	0.79/0.96/0.95	0.673
DIRECT	additive routing	**0.636**	[0.621, 0.651]	0.927/0.873/0.858	**0.81**/0.96/0.95	**0.681**

Table 3. Bottleneck necessity for DIRECT (Dice, n=40). Each path is ablated separately; *zero_routing* is the direct-mode number. Ablating only the decoder's bottleneck input would leave the routing term intact and wrongly report the bottleneck as unnecessary.

	base	bypass	zero_expert	**zero_routing**
NCR	0.483	0.484	0.053	**0.006**
Edema	0.898	0.477	0.333	**0.418**
ET	0.852	0.597	0.428	**0.172**

$[+0.609, +0.641]$ with Wilcoxon p=2.6×10^{-29} over 168 matched cases. Fig. 2 shows the same result as images.

Two independent measurements corroborate it. Removing only the routing term collapses the model (Table 3): NCR −99%, ET −80%, edema −53%, against ≈1.5% for zeroing V0's entire bottleneck. And the magnitude ratio ρ converged at 0.35, so the routing term dominates the logits roughly 3 : 1—the failure mode we instrumented for did not occur.

The price. Whole tumor is free (0.9273 vs. 0.9272); TC and ET cost ≈1 Dice point each. Alignment is marginally *better* than the inert baseline (CAS_{fg} NCR 0.809 vs. 0.802), calibration improves substantially (ECE 0.841 → 0.681), and token-level AUROC rises on edema (0.638 → 0.900) and ET (0.622 → 0.839) while falling on NCR (0.937 → 0.783).

Alignment emerges from the task. One consequence deserves emphasis. Because the routing logits are part of the output, $\mathcal{L}_{\text{seg}}$ itself trains the routing to be concept-aligned. By epoch 10, with $\mathcal{L}_{\text{align}}$ still switched off, CAS_{fg} has reached 0.41/0.42 on edema and ET, where V0 remains at the floor (0.08/0.12). This directly answers the objection that CAS_{fg} only measures what $\mathcal{L}_{\text{align}}$ trains: here alignment appears without it.

The effect is specific and local. A large flip rate would mean little if the intervention simply perturbed everything, so we report the two controls that separate concept-specific control from generic disturbance (Table 4). Classes other than the source and target move by 0.079 on average—an order of magnitude below the 0.636 target effect—and voxels *outside* the re-routed region move by 0.005. The source class falls by 0.544, mirroring the target's rise. The intervention does what it is supposed to do, where it is supposed to do it.

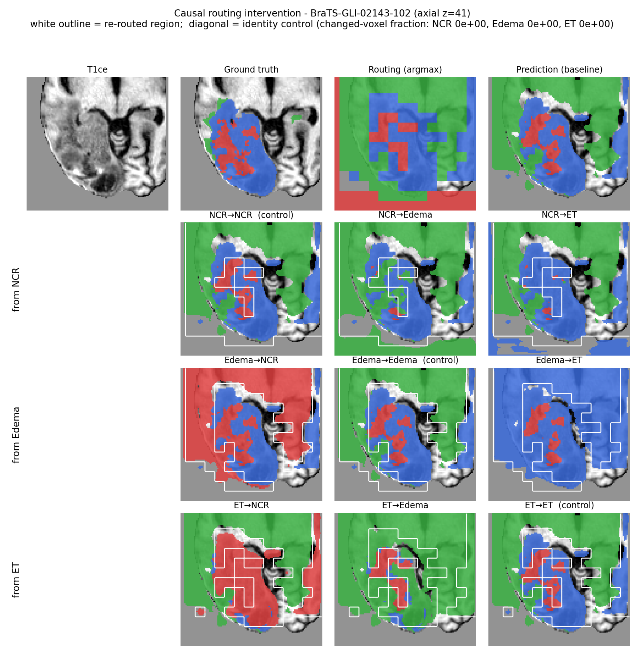

Figure 2. **Causal intervention, rendered.** Top: input, ground truth, routing argmax, baseline prediction. Grid: rows are the *source* concept, columns the *target*; each cell is the segmentation after forcing every token the router assigned to the row concept through the column concept's expert, with the re-routed region outlined in white. The diagonal is the identity control and is pixel-identical to the baseline (changed-voxel fraction 0 for all three), so the off-diagonal changes are attributable to re-routing alone.

Scope matters, and the smaller number is the honest one. A router typically assigns most of the volume to background: here roughly a quarter of all tokens go to the edema expert, the majority of them normal tissue. Re-routing *every* token of a concept therefore repaints the whole brain, and reports 0.816 rather than 0.636. That larger figure is not wrong, but it mostly demonstrates that

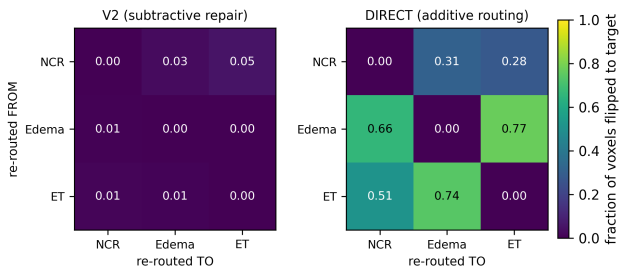

Figure 3. **Causal control, V2 vs. DIRECT**, on a shared colour scale. Rows are the source concept, columns the target; each cell is the fraction of re-routed voxels whose predicted label became the target. Subtractive repair leaves the matrix uniformly dark; additive routing does not. Diagonals are the identity control and are exactly zero in both.

Table 4. Specificity and locality of the intervention (DIRECT, n=168, mean over off-diagonal concept pairs). Restricting the intervention to the tumor neighborhood does not merely lower the headline—it makes the effect *cleaner* on both controls.

	tumor-restricted	all tokens
target class ↑	$\mathbf{0.540}$	0.798
source class ↓	-0.544	-0.514
off-target classes → 0	$\mathbf{0.079}$	0.151
outside the region → 0	$\mathbf{0.005}$	0.016

the output follows the routing everywhere, which under Eq. 3 is close to tautological. Restricting to the tumor neighborhood measures concept control where it is clinically meaningful—and, as Table 4 shows, roughly halves the off-target disturbance and cuts the out-of-region leakage by $3.5\times$. We quote the restricted number throughout and report the scope with every measurement.

Where it is weakest, and why. The effect is not uniform, and the asymmetry is mechanistically informative rather than merely a caveat. Necrotic core is weak as a *source* (0.28–0.31) but works as a *target* (0.51–0.66): the model can be pushed into calling a region necrosis, but pushing necrosis tokens elsewhere works less well. The source-class posteriors say why. Routing edema or ET tokens away drops their own class by 0.81 and 0.58 respectively, whereas routing necrosis away drops it by only 0.23—because the model was never confident about necrosis to begin with (its isolated Dice is 0.483, against 0.898 for edema). There is little confidence to remove, so the re-routed region is contested rather than reassigned, and off-target classes absorb the difference (0.09–0.11 from necrosis against 0.03–0.06 elsewhere). Causal control is therefore bounded by how decisively the model held the original concept, which is a property of the underlying segmentation problem rather than of the routing mechanism. Necrotic core is the scarcest

Table 5. The diagnostic protocol across configurations. Alignment separates none of them; the causal tests separate DIRECT from both. Ablation reports edema Dice with the module's contribution removed, relative to that configuration's own baseline.

	V0	V2	**DIRECT**
alignment — cannot separate them			
$\mathrm{CAS}_{\mathrm{fg}}$ (edema)	0.962	0.961	0.962
concept-discrim.	0.883	0.879	0.881
causal — separates DIRECT			
T1 intervention ↑	0.001	0.011	$\mathbf{0.636}$
T2 ablation, edema ↓	-1.6%	-0.7%	$\mathbf{-53}\%$
T3 probe vs. router	0.839 vs. 0.872; 0.811 unsup.		

concept at a 16^3 bottleneck—tens of tokens per case—and is the consistent weak spot across every metric we report, not only this one.

5.7. The protocol, applied

Table 5 collects the three tests across the three configurations. Read down a column and the story of each design is visible in three numbers; read across a row and the tests disagree with the alignment metrics in exactly the way Sec. 4 predicts. V0 and V2 are indistinguishable on every causal test while differing on none of the alignment ones, which is the whole point: alignment cannot separate them, and intervention can.

5.8. Backbone transfer

The natural design behaves the same on both backbones. A Swin transformer [6, 13] with a UNETR decoder [8] reaches Dice $0.915/0.845/0.837$, $\mathrm{CAS}_{\mathrm{fg}}$ $0.69/0.94/0.93$ and discriminability 0.872; the SegResNet CNN reaches $0.927/0.883/0.869$, $0.80/0.96/0.95$ and 0.883. They differ in pretraining, capacity and accuracy, yet both align well and both are inert. The leaks are properties of the architecture family rather than of one backbone, and the transformer shows the placement failure of Sec. 3.2 in its sharpest form: its module sits at stage 2 of 5, so the two deeper and more semantic encoder stages bypass it entirely.

6. Discussion and Limitations

What alignment metrics can and cannot show. $\mathrm{CAS}_{\mathrm{fg}}$ and concept-discriminability measure correlation between an internal signal and annotated concepts. A module can maximize both while contributing nothing to the prediction, and the alignment loss trains precisely what these metrics score. Any claim that a concept layer makes a model interpretable requires an intervention. We state this as the paper's transferable lesson because it cost us: on the correlational evidence alone we had concluded the module worked, and only the causal test reversed that reading.

Additive beats subtractive. Both failed repairs share a shape: they try to force a decoder that segments well without its bottleneck to depend on it, by removing alternatives. The optimizer routed around both. Making the routing a term of the output instead means causal influence is arithmetic, and no weight configuration undoes it after the fact. We expect this to generalize to other concept-bottleneck designs in dense prediction, though we have tested it only here.

Reported metrics that do not transfer. Three comparisons become invalid under Eq. 3 and we flag them rather than tabulate them. Decoder Divergence, $1 - \mathrm{Pearson}(\mathrm{decoder}, \mathrm{routing})$, is circular: the decoder output *contains* the routing term by construction, so it measures how much the refinement perturbs the routing, not independent agreement. Segmentation loss is not comparable across additive and non-additive configurations, because the larger logit scale lowers cross-entropy without improving the argmax. And V2's low off-target figure (0.022 vs. DIRECT's 0.079) reflects its near-zero effect rather than better precision—an inert module perturbs nothing, including the things it should.

No controlled accuracy comparison. We did not train plain-backbone counterparts under our split, so we make *no* claim that the module is accuracy-neutral in absolute terms. The $\approx$1 Dice point in Table 2 is a *relative* cost measured against the same architecture with an inert bottleneck, which is the quantity the trade-off argument needs, but it is not a statement about the cost of adding the module to a plain network.

An unexplained interaction. Zeroing both CCR paths is *less* damaging than zeroing the routing term alone (edema 0.801 vs. 0.418, Table 3). The decoder is trained to add to a routing base; stripping the base while leaving the expert path appears to leave it miscalibrated, whereas zeroing both puts it in a cleaner regime. We report this because it is real and we cannot yet explain it.

Resolution, granularity, and scope. The bottleneck is 16^3, so the explanation is coarse and necrotic core—tens of tokens per case—is the consistent weak spot across every metric. $K{=}4$ follows the BraTS protocol and forces transition-zone tokens into one concept. We evaluate one dataset and one anatomy; the two leaks are properties of the architecture family rather than of the data, but that expectation is untested. Routing entropy is a per-voxel uncertainty signal but poorly calibrated (ECE 0.68), and plain segmentation confidence predicts model error better than any routing-derived signal we tried (0.883 vs. 0.786), so we make no uncertainty claim.

What the explanation is, and is not. CCR explains at the level of clinical concepts—which expert processed each region, and with what confidence—not the router's internal reasoning. Its advantage over a saliency map is that it is concept-labeled, intrinsic, modular, and now demonstrably causal; it is not spatially sharper, and the segmentation output itself remains a finer-grained concept map. What the routing adds is a low-dimensional, inspectable, and *manipulable* account of the concept decision that the output does not provide about itself.

What a causal bottleneck offers, and reproducibility. We ran no reader study, so we make no claim that clinicians want concept-level explanations. What the result changes is the kind of question the model can answer: a saliency map supports only "where did the model look", an aligned but inert bottleneck adds "what did the model call this region" (which the output already answers at higher resolution), and a *causal* bottleneck additionally supports counterfactuals—"what if this region were edema—answered by the model itself. Before the repair that question was unanswerable in principle, because the output did not depend on the concept assignment. All configurations differ only in the switches of Sec. 3.3; backbone, losses, curriculum, optimizer, split and seed are shared, so the comparisons isolate the architectural change. Every number is computed by the released protocol scripts from committed per-case files, and the intervention diagonal is recomputed on every run as a self-check.

Practical guidance. Three recommendations follow, for anyone placing a concept layer in a dense-prediction network. Run the intervention before reporting alignment—it needs no retraining and was the only test that separated a working design from a decorative one. Check where the bottleneck actually is: in a hierarchical encoder the deepest stage is not the one whose width matches the module. And prefer additive to subtractive coupling, since both subtractive repairs we tried were absorbed by the optimizer.

7. Conclusion

We set out to make the routing decision the explanation, and found that the obvious implementation yields a concept bottleneck that is beautifully aligned and causally inert: $\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96, best-in-class concept discriminability, and a prediction that does not move when the routing is overwritten. The cause is not degenerate experts but two architectural leaks alignment metrics cannot see. Making the routing an additive term of the output raises causal control from 0.011 to 0.636 for $\approx$1 Dice point. The lesson generalizes beyond this system: **a concept layer is an explanation only if the model consults it, and demonstrating that requires intervening, not correlating.**

References

[1] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *ACL*, 2020. 2

[2] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *NeurIPS*, 2018. 2

[3] Ujjwal Baid et al. The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification. *arXiv:2107.02314*, 2021. 4

[4] David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. Network dissection: Quantifying interpretability of deep visual representations. In *CVPR*, 2017. 2

[5] Chaofan Chen, Oscar Li, Daniel Tao, Alina Barnett, Cynthia Rudin, and Jonathan K Su. This looks like that: deep learning for interpretable image recognition. In *NeurIPS*, 2019. 2

[6] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021. 7

[7] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *JMLR*, 23:1–39, 2022. 2

[8] Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger R Roth, and Daguang Xu. Unetr: Transformers for 3d medical image segmentation. In *WACV*, 2022. 7

[9] Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. 3

[10] Sarthak Jain and Byron C Wallace. Attention is not explanation. In *NAACL-HLT*, 2019. 2

[11] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). In *ICML*, 2018. 2

[12] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *ICML*, pages 5338–5348, 2020. 1, 2

[13] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021. 7

[14] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *NeurIPS*, 2017. 1, 2

[15] Bjoern H Menze et al. The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE TMI*, 34(10): 1993–2024, 2014. 4

[16] Carlos Riquelme et al. Scaling vision with sparse mixture of experts. In *NeurIPS*, 2021. 2

[17] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015. 3

[18] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019. 1

[19] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *ICCV*, pages 618–626, 2017. 1, 2

[20] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *ICLR*, 2017. 2

[21] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *ICML*, pages 3319–3328, 2017. 1, 2